Sanandan Ojha

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RESEARCH SUMMARY

Biomedical engineering and mathematics undergraduate (B.S., May 2027) with hands-on experience collating, cleaning, and analyzing scientific and clinical datasets in Python and R, including work on the HiPerGator Linux cluster. Background spans AI-driven protein binder design (BindCraft, AlphaFold), machine-learning-based medical image synthesis, and clinical imaging data analysis in radiopharmaceutical therapy. Eager to support drug-discovery machine learning by building well-organized, documented, and reproducible datasets.

EDUCATION

University of Florida, Gainesville FL: 3.91 GPA *August 2023—May 2027*

Bachelor of Science, Mathematics
Bachelor of Science, Biomedical Engineering
Minor, Statistics

RESEARCH EXPERIENCES

University of Florida, Honors Thesis

Advisors: Dr. Stepien, Dr. Murfee, Dr. Mancy *August 2026—Present*

- Implementing multiple inlet-outlet system for circulatory system code base to identify shear stress in capillaries.
- Using shear stress data to estimate hepatic clearance under various scenarios.

University of Florida, Mathematical Biology

Advisor: Dr. Stepien *October 2025—Present*

- Developing a **2**-layer continuum mechanics model for cell migration in gastrulation stage of development.
- Using **UQ-Lab** and MATLAB to conduct Sensitivity Analysis and Approximate Bayesian Computation – Sequential Monte Carlo (**ABC-SMC**) to fit parameters and models to experimental data on HiPerGator.

University of Florida, Systems Medicine

Advisor: Dr. Brunson *May 2026—Present*

- Developing a novel mechanistic model for primary graft dysfunction in lung transplantations.
- Utilizing pathway mining and conceptual modeling to develop the model.

Florida Theragnostic Center

Advisor: Dr. Krishnan *May 2026—May 2026*

- Case study on Xofigo (^{223}Ra Dichloride) imaging and therapy in patients.
- Demonstrated a correlation between Bone Scan Index and patient outcomes.

University of Florida, International Genetic Engineering Machine (iGEM)

Advisor: Dr. Pan *February 2025—October 2025*

- Utilized AI to design a novel binder targeting osteocalcin, including **BindCraft** workflow on HiPerGator.
- Contributed to project documentation using **React** to receive **Silver** recognition at 2025 Grand Jamboree.

Advisor: Dr. Pan *February 2024—October 2024*

- Adapted a differential model of sepsis in bone marrow into an agent-based model using **NetLogo.**
- Contributed to project documentation using HTML to receive **Gold** recognition at 2024 Grand Jamboree.

Indiana University Indianapolis, Research Experience for Undergraduates

Advisor: Dr. Zhu *May 2025—July 2025*

- Developed a Lattice-Boltzmann Immersed-Boundary (**LB-IB**) method for cells in low Reynolds number flows.

- Increased efficiency by **300 steps/sec** with minimal error **1E-9 per timestep** over traditional LB-IB.
- Constructed an implicit LB-IB framework for cell migration in low Reynolds number scenarios.

University of Florida, Dream Team Engineering
Advisor: Dr. Shah *January 2025—Present*
- Testing Liver, Kidney, Cholecystectomy, and Caudal Block **surgical simulators** using medical residents/students.
- Gathering data to establish the efficacy of simple surgical models in giving residents hands-on experience.

Advisor: Dr. Wong *August 2024—May 2025*
- Using **Diffusion ML models** to generate MRI images from CTs for perioperative planning of patients undergoing Deep Brain Stimulation and contraindicated for MRIs.

Advisor: Dr. Chandran *September 2023—January 2025*
- Constructed **experimental design** for efficacy of a 3D printed Berlin Heart and supraventricular tachycardia patient education models.

University of Florida, Biochemistry
Advisor: Dr. Caglayan *January 2024—May 2025*
- **Overexpression** and **purification** of recombinant human DNA repair proteins.
- Biochemical and structural characterization of DNA repair proteins.

University of Pittsburgh, Radiology
Advisor: Dr. Muthukrishnan *May 2022—June 2022*
- Collated clinical and imaging data to evaluate Prostate Specific Membrane Antigen (PSMA) imaging efficacy.
- Established a correlation between patient outcomes and PSMA PET performance relative to conventional PET.

WORKSHOPS AND ENRICHMENT

University of Arizona, Preliminary Arizona Winter School (PAWS) *September 14, 2026—November 10, 2026*
Coordinator: Dr. Brandon Levin
- Attended lectures on Quaternion Algebra by Dr. Jen Berg.
- Weekly problem set completed in collaboration with sessions led by Dr. Divyasree C-Ramachandran.

University of Florida, Directed Reading Program *August 31, 2026—Present*
Coordinator: Dr. Sara Pollock
- Reading "Weak Convergence Methods for Non-linear Partial Differential Equations" by Evans.
- Attending weekly guided session by Dylan Connell, PhD candidate.

PI3, Polyploidy Undergraduate Research Experience *June 29—July 1, 2026*
Coordinator: Dr. Pam Soltis
- Attended lectures on transcription analysis, proteomics analysis, and polyploidy investigation.

Texas A&M, Modeling and Simulating with Partial Differential Equations (PDEs) *May 18—May 29, 2026*
Coordinator: Dr. Matthias Maier
- Attended lectures on PDE theory, Finite Difference and Finite Element methods, and Transport Equations.
- Created a solver for the Radiation Transport Equation directed by Dr. Guermond.

University of Florida, Infectious Diseases and Economic Growth *March 26—March 29, 2026*
Coordinator: Dr. Calistus Ngonghala
- Attended lectures on modeling infectious disease spread and interactions with the economy.
- Designed a model for early COVID-19 spread in Florida, including parameter estimation using ABC.

POSTER PRESENTATIONS

Ojha, Sanandan; *Dadlani, Rishi; Divan, Ryan; and Ratcliff, Gavin (November 2026). Radiation Transport Equation. Poster presented at 2026 UIC Undergraduate Mathematics Symposium, Chicago, IL.*

Ojha, Sanandan*; Kulkarni, Pranav; Patel, Sharad; Lizdas, Dave; Mesber, Simon; Lampotang, Samsun; and Dooley, Cole (November 2026). Efficacy of a Mixed-Reality Simulator for Ultrasound-Guided Infant Caudal Epidural Block Training: Study Design. Poster presented at 2026 Fall Undergraduate Research Symposium, Gainesville, FL.*

Ojha, Sanandan*; Kulkarni, Pranav; Patel, Sharad; Lizdas, Dave; Mesber, Simon; Lampotang, Samsun; and Dooley, Cole (October 2026). Efficacy of a Mixed-Reality Simulator for Ultrasound-Guided Infant Caudal Epidural Block Training: Study Design. Poster presented at 2026 Biomedical Engineering Society Annual Meeting, Orlando, FL.*

Ojha, Sanandan*; Nguyen, Thien-Nhi; Shreekanth, Shreya; and Shah, Rushi (October 2026). Evaluation of a Kidney Transplant Model for Educational Skill Development. Poster presented at Biomedical Engineering Society Annual 2026 Meeting, Orlando, FL.*

Ojha, Sanandan *& Stepien, Tracy (April 2026). Mathematical Modeling of Intercalation in Two Cell Layers During the Gastrulation Stage of Embryonic Development. Poster presented at 2026 Spring Undergraduate Research Symposium, Gainesville, FL.*

Ojha, Sanandan *& Stepien, Tracy (March 2026). Mathematical Modeling of Intercalation in Two Cell Layers During the Gastrulation Stage of Embryonic Development. Poster presented at 2026 Infectious Diseases and Economic Growth Workshop, Gainesville, FL.*

Ojha, Sanandan*; Gill, Jali; and Zhu, Luoding (July 2025). Modeling Cell Migration with a Linearized Lattice-Boltzmann Immersed-Boundary Method. Poster presented at 2025 Indiana University Indianapolis Summer Research Symposium, Indianapolis, IN.*

Ojha, Sanandan*; Kennedy, Hannah; Sharma, Utsav; and Labasbas, Joshua (April 2025). Synthetic MRI Generation for Deep Brain Stimulation Preoperative Planning. Poster presented at 2025 Yale Undergraduate Research Conference in New Haven, CT.*

Ojha, Sanandan*; Kennedy, Hannah; Sharma, Utsav; Jain, Ayush; and Kelegama, Ania (November 2023). 3D Modeling of Berlin Heart Devices: Mitigating Pediatric Anxiety in Congenital Heart Disease. Poster presented at 2023 Fall Undergraduate Research Symposium by Center for Undergraduate Research, Gainesville, FL.*

ORAL PRESENTATIONS

Ojha, Sanandan; *Suarez, Santiago; and Connell, Dylan (November 2026). Weak Convergence Methods for Partial Differential Equations. Presented at 2026 UF DRP Mathematics Symposium, Gainesville, Fl.*

Ojha, Sanandan; *Dadlani, Rishi; Divan, Ryan; and Ratcliff, Gavin (May 2026). Radiation Transport Equation. Presented at 2026 TAMU Undergraduate Summer School in Modeling and Simulating with PDEs, College Station, TX.*

Ojha, Sanandan *& Stepien, Tracy (April 2026). Mathematical Modeling of Intercalation in Two Cell Layers During the Gastrulation Stage of Embryonic Development. Presented at 2026 Undergraduate Mathematics Research Symposium, Gainesville, FL.*

Ojha, Sanandan*; Kulkarni, Pranav; and Patel, Sharad (April 2026). Modeling AI Model Collapse Using Markov Chains and Layered Discrete Distributions. Presented at 2026 Undergraduate Mathematics Research Symposium, Gainesville, FL.*

Ojha, Sanandan*; Gill, Jali; and Zhu, Luoding (July 2025). Modeling Cell Migration with a Linearized Lattice-Boltzmann Immersed-Boundary Method. Presented at 2025 Indiana Undergraduate Math Conference, Indianapolis, IN.*

Ojha, Sanandan *& Muthukrishnan, Ashok (October 2022). Utility of SSTR PET Imaging in Management of Olfactory Neuroblastomas. Presented at 2022 Annual Southeastern Chapter of Society of Nuclear Medicine and Molecular Imaging, Savannah, GA.*

PUBLICATIONS

Ojha, Sanandan & Muthukrishnan, Ashok *(July 2022). Pluvicto Approval Launches Theragnostic in Prostate Cancer. Targeted Therapies in Oncology. 11(10):77.*

Manuscripts in Preparation

***Sanandan Ojha**, Jean-Luc Choiseul, Holley E. Lynch, Melis Hazar, Lance A. Davidson, Tracy L. Stepien. Model Selection of Collective Cell Migration in Multi-Layer Aggregates.* In Preparation.

*Luoding Zhu, **Sanandan Ojha,** Jali Gill, Jared Barber. An Implicit Lattice Boltzmann – Immersed Boundary Method for Cell Migration.* In Preparation.

HONORS

University Scholars Program, CLAS, Total $1750 — *Fall '26—Spring '27*
Best Student Presenter, SECSNMMI — *Fall '22*
Outstanding SCUDEM X Participant, SIMIODE — *Fall '25*
Gold Medal, IGEM — *Fall '24*
Silver Medal, IGEM — *Fall '25*
Successful Participant MCM/ICM, COMAP — *Spring '24, Spring '25, Spring '26*
Successful Participant SCUDEM, SIMIODE — *Fall '24*
UF Math Circle Undergraduate Leader, UF Mathematics — *Spring '26*
Undergraduate Research Program, UF Mathematics, $600 — *Spring '26*
BME LA Program, College of Engineering, Total $1500 — *Spring '25, Spring'26*
Dean's List, CLAS — *(3 awards) Spring '25—Spring '26*
Dean's List, College of Engineering — *(7 awards) Fall '23—Spring '26*

PROJECTS

Senior Design (Sponsor: CONMED)
Distal Bicep Tendon Repair System — *August 2026—Present*

- Designing a proof-of-concept distal bicep tendon repair system for our industry sponsor, CONMED, with a team of eight senior engineering students.
- System includes implant, instruments, and sterile packaging; a sports medicine surgeon will evaluate the design.

Quantitative Physiology: BME4409
Restenosis in Stents — *August 2026—December 2026*

- Adapting a model for restenosis built for coronary stents to self-expanding nitinol stents in the leg.
- Simplified the model to 2D by applying a constant outward pressure to represent the load on the femoral artery.

Modeling and Simulating with Partial Differential Equations
Radiation Transport Equation — *May 18—May 29, 2026*

- Developed theory and solver for the Boltzmann Equation of Inertial Confinement Fusion using 1D slab geometry.
- Coupled Streamline Upwind Petrov-Galerkin solver with Richardson Iteration for integro-differential term.
- Solved benchmark problems including the Reed problem and the Grazing problem.

SIMIODE Challenge Using Differential Equations Modeling
Problem A: "Pediatric Prosthetics" — *October 16—November 10, 2026*

- Constructed a 2D quasi-static model for patient growth and motion paths using inverse kinematics and forces.
- Made 10-minute video on result, awaiting judging.

Problem B: "An AI Ouroboros" — *October 18—November 11, 2025*

- Constructed a modified Markov Chain and layered discrete distributions to model AI collapse in a task space.
- Made 10-minute video on results, recognized as **Outstanding Participant** and featured on MathWorks website.

Mathematical Contest in Modeling/Interdisciplinary Contest in Modeling
Problem A: "Modeling Smartphone Battery Drain" — *January 29—February 2, 2026*

- Designed **continuous time models** in MATLAB and used **UQ-Lab** for sensitivity analysis.
- Wrote 25-page report on model, recognized as a **Successful Participant**.

Problem A: "The Constant Wear on Stairs" — *January 23—January 27, 2025*

- Created **3 difference equations** to estimate 3 values using limited, measurable parameters.
- Wrote 15-page report on models and was recognized as a **Successful Participant**.

Problem C: "Momentum in Tennis" *February 1—February 5, 2024*

- Analyzed a data set from the 2024 Wimbledon Championships **to construct** 4 models using **R-Programming**.
- Created a 25-page report describing analysis, findings, and models, recognized as a **Successful Participant**.

Infectious Diseases and Economic Growth: Modeling Early COVID-19 in FL *March 26—March 29, 2026*

- Created a simplified **SIRS** model with economic interactions in **Python**, fitted against early Florida infection data.
- Used **Approximate Bayesian Computation** to reduce parameter space and compare likely long-term behavior.

PCB3713C Auto-Sorter *September 2025—October 2025*

- Created an executable to format student submissions for uploading to Canvas using **Python**.

Dream Team Engineering Bias Detection *August 2025—September 2025*

- Analyzed 8 cycles of past application data and acceptances for Dream Team Engineering using **R Programming**.
- Looked for bias and trends in individual reviewer decisions and overall application pool.

Herbert Wertheim College of Engineering Outreach Design Competition *February 17, 2026*

- Designed 3D printed mobile cranes with hydraulic booms for 6th-8th graders using **Fusion 360**.
- Created matching handouts to illustrate concepts relating to Pascal's Law and Mechanical Advantage to students.

Dream Team Engineering Designathon

Multi-Purpose Piston-Powered Walker *November 7—November 9, 2025*

- Designed a modified walker that provides powered assistance to the user in traversing stairs and getting seated.
- A 3D CAD of the walker was created using **Fusion 360**, along with budget and stakeholder analysis.

Prescription Puzzles *October 12—October 13, 2023*

- Created a low-cost pill dispenser using a single **Arduino** that is activated by completion of a Jigsaw puzzle.

UCSAS USOPC Data Challenge *November 1, 2023—January 15, 2024*

- Chose 10 athletes for US Olympic Gymnastics team by analyzing collated historical event data in Python.
- Wrote a short report detailing methodology and conclusions.

TEACHING EXPERIENCES

Undergraduate Teaching Assistant

CHM2100: Organic Chemistry *May 2026—August 2026*

- Created a Discord environment for students to engage TAs and study, with over **100** students participating.
- Held weekly review sessions covering material from the week as well as test practice.

PCB3713C: Cellular and Systems Physiology *August 2025—August 2026*

- Designed new labs and activities to accommodate expanded class times for Fall 2026.
- Ran weekly student labs, held office hours, provided mentorship to student groups, and graded assignments.

BME Learning Assistant

BME3053C: Computer Applications for BME *February 2026—May 2026*

- Hosted office hours, helped students with homework, and provided mentorship as a peer mentor.

BME3060: Biomedical Engineering Fundamentals *February 2025—May 2025*

- Hosted office hours, created exam reviews, and provided mentorship as a peer mentor.

LEADERSHIP

MATLAB Club *August 2026—Present*

- Hold events promoting the use of MATLAB for students, including an info session for the SCUDEM competition.

Dream Team Engineering

Director of Research *May 2025—May 2026*

- Directly oversaw and managed **5** undergraduate research teams with **34** members for the academic year.
- Evolved 'Research Bootcamp' to showcase fundamentals of ANOVA analysis and Machine Learning principles.

Assistant Director of Research *August 2024—May 2025*

- Directly oversaw and managed **4** undergraduate research teams with **29** members for the academic year.
- Ideated and held 'Research Bootcamp' to introduce fundamentals of research and scientific writing.

Badminton Club

Executive Board Member *August 2025—Present*

- Organized volunteer events for the UF Badminton Club, creating community and helping the larger Alachua area.
- Safety Officer in charge of first response of health emergencies at club practices, tournaments, and other events.

BME Inclusive Excellence

Secretary *August 2024—May 2025*

- Increased undergraduate participation in BME inclusive excellence events, fostering greater community in BME.
- Designed an event for World Mental Health Day, highlighting mental health among academics and students.

STEM OUTREACH

University of Florida, Math Circle *August 2025—Present*

- Helping elementary - high school students learn problem-solving skills by tackling math games and problems.
- Designed and hosted Math Circle meeting about teaching Monte Carlo estimation and probability.

University of Florida, MentorGNV *August 2025—May 2026*

- Helping elementary/middle school students in after/during school activities, including tutoring in prealgebra.

Event Volunteering

University of Florida, Boo at the Zoo *October 28, 2026*
University of Florida, Math Festival *March 29, 2026*
Alachua, STEAM Night *March 27, 2026*
Indiana University Indianapolis, STEM YES Engineering/Technology Event *June 18, 2025*
Alachua, Regional STEM Fair *January 5, 2025*
University of Florida, Florida Science Olympiad State Event *April 6, 2024*

COMMUNITY SERVICE

University of Florida, Badminton Club

Jimmy Carter Invitational *2025—2026*
NAIA Track and Field *2025—2026*
Keep Alachua County Beautiful *2024—2026*
Food4Kids *2026*
Gator Slam *2023—2025*
Gator Smash *2024—2026*

CLINICAL EXPERIENCE

West Palm Beach, FL, Florida Theragnostic Center *May 2026—August 2026*

Supervisor: Dr. Krishnan

- Shadowed nurses, technicians, and radiologists throughout workday.

West Palm Beach, FL, VA Medical Center *June 2022—August 2022*

Supervisor: Dr. Mullin

- Shadowed multiple radiologists throughout workday.

Supervisor: Dr. Shetty

- Shadowed oncologist, dosimetrist, and physicist throughout workday.

SKILLS

Languages: Python, MATLAB, R-Programming, React, Java, LaTeX/Markdown, SQL (in progress).

Technical Skills: NetLogo, Arduino IDE, KiCad, ImageJ, UQ-Lab, BindCraft, AlphaFold, Linux/HPC (HiperGator).

Wet-Lab: Cell Culturing, Protein Purification, Bacterial Transformation, Overexpression, SDS-PAGE.

Modeling: Continuum Mechanics, Lattice Boltzmann–Immersed Boundary, Agent-Based Modeling, Approximate Bayesian Computation.

Certification: American Red Cross Basic Life Support